

N.B.:

1. Figures shown in the right margin indicate full marks.
2. Answer **Six** questions, taking **Three** from each section including Q.1, Q.2, Q.5 and Q.6
3. Use separate answer script for each section.
4. Assume reasonable data if necessary.

Section-A

- Q.1** (a) For the **Figure Q. 1(a)**, sketch a free body of the members designated by the letter A and B. All plane surfaces and pins are smooth (frictionless). Flexible, weightless cords are represented by single lines. [CLO1, PLO1, C2] 5

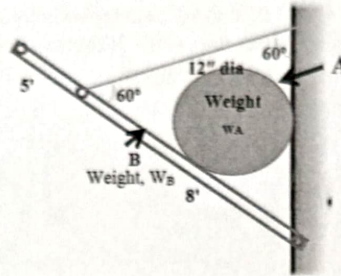


Figure Q. 1(a)

- (b) Determine the tension developed in wires CA and CB required for equilibrium of the 10 kg cylinder. Take, $\theta = 40^\circ$. [CLO2, PLO2, C3] 5

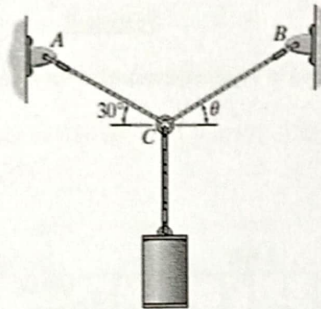


Figure: Q.1 (b)

- Q.2** (a) A flexible chord is loaded with uniformly horizontally. Derive the expressions for maximum support tension. [CLO1, PLO1, C2] 5
- (b) A cable weighing 6 lb per ft is suspended from two supports on the same level. (a) For a span of 800 ft and a sag of 60 ft, determine the maximum tension. (b) If six cables as described pass across a supporting tower and if the span and sag are the same on each side of the tower, what vertical load must the tower carry? Assume the curve to be parabolic. [CLO2, PLO2, C4] 5
- Q.3** (a) A body of weight 300 N is lying on a rough horizontal plane having a coefficient of friction as 0.3. Find the magnitude of the force, which can move the body, while acting at an angle of 25° with the horizontal. [CLO2, PLO2, C4] 4

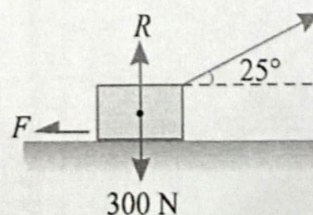


Figure: Q.3 (a)

- (b) Determine the maximum load (W) which a force P equal to 5 kN will hold up, if the coefficient of friction at C is 0.2 in the arrangement shown in **Figure: Q.3 (b)**. Neglect other friction and weight of the member. [CLO2, PLO2, C3] 6

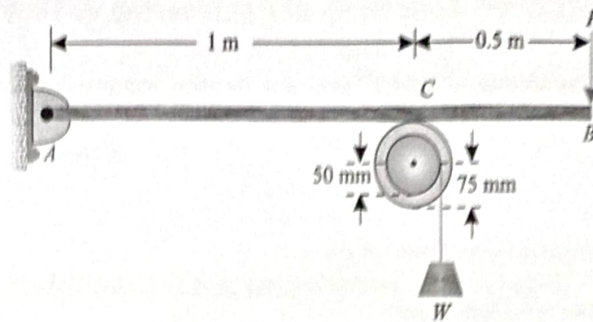


Figure: Q.3 (b)

- Q.4 (a) If the motion of a parachutist (before the parachute opens) is expressed by a relation $s = ct^2$ ft where c is constant. Determine his speed and acceleration 2 sec after he leaves the plane, if $c = 15$. Assume that he moves in vertical path. [CLO2, PLO2, C4] 4
- (b) Three balls A, B and C of weight 30N, 50N and 80N are moving in the same direction with velocities 10 m/s, 5 m/s and 2 m/s respectively. If the ball A strikes with the ball B which in turns strikes with the ball C. If the coefficient of restitution is 0.8. Find, 6
- Velocities of the ball after impact.
 - Loss of K.E. during impact.
- [CLO2, PLO2, C3]

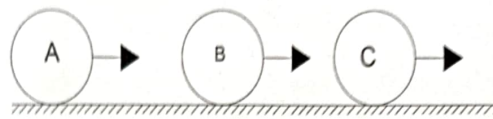


Figure Q. 4 (b)

Section-B

- Q.5 (a) A beam is loaded as shown in Figure Q.5.(a). Find the support reactions at A and C. 5
- [CLO3, PLO2, C4]

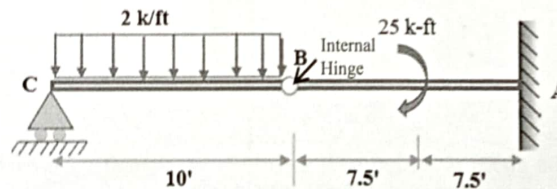


Figure Q. 5(a)

- (b) Calculate the support reactions of the following frame shown in Figure Q. 5(b). 5
- [CLO2, PLO2, C4]

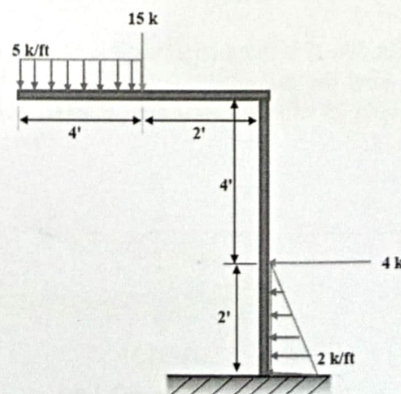


Figure Q. 5(b).

Find out the center of gravity of the object as shown in **Figure: Q.6 (a)**.

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[CLO3, PLO2, C4]

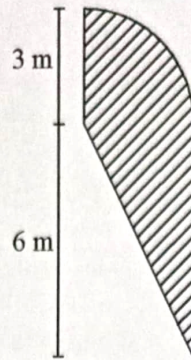


Figure: Q.6 (a)

(b) Determine the Centroid of the following section as shown in **Figure: Q.6 (b)**.

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[CLO3, PLO2, C4]

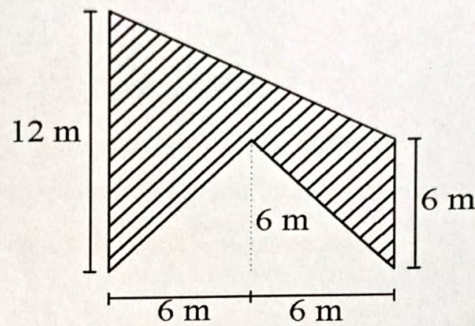


Figure: Q.6 (b)

Q.7 (a) Determine the member force of DE, BE, GH, and FH of the truss as shown in **Figure: Q.7 (a)**. [CLO3, PLO2, C3]

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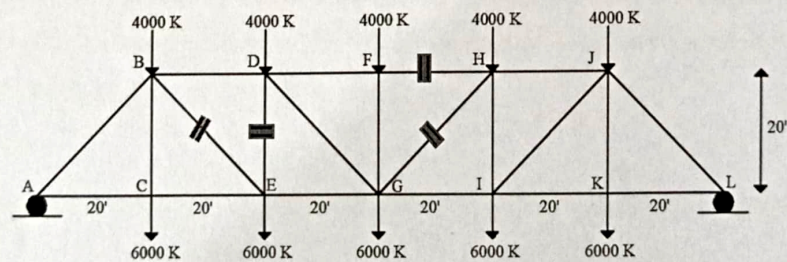


Figure: Q.7 (a)

(b) Determine the support reactions and the member force of BE, AB, BF and FG of the truss as shown in **Figure: Q.7 (b)**. [CLO3, PLO2, C3]

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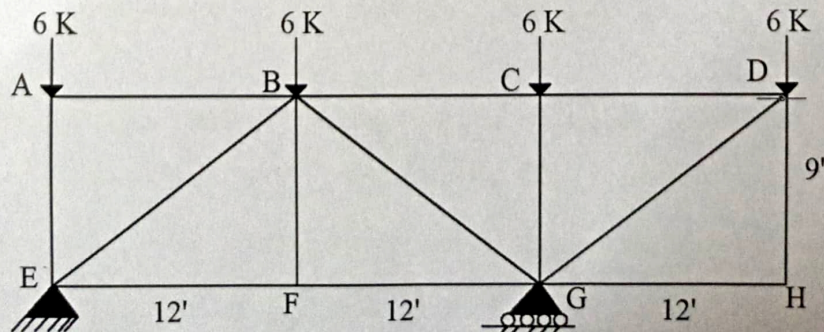


Figure: Q.7 (b)

- Q.8** (a) Find the moment of inertia of the section shown in **Figure: Q.8(a)** about vertical axis passing through the centroid of the section. [CLO3, PLO2, C3]

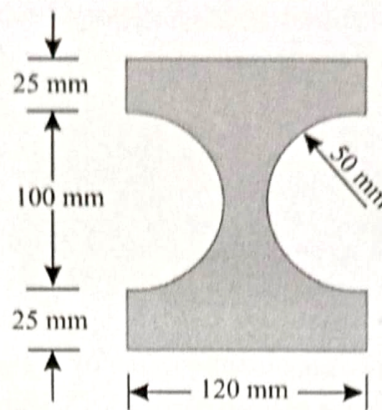


Figure: Q.8(a)

- (b) Compute the moment of inertia of the section shown in **Figure Q. 8 (b)** about axis K-K. [CLO3, PLO2, C3] 5

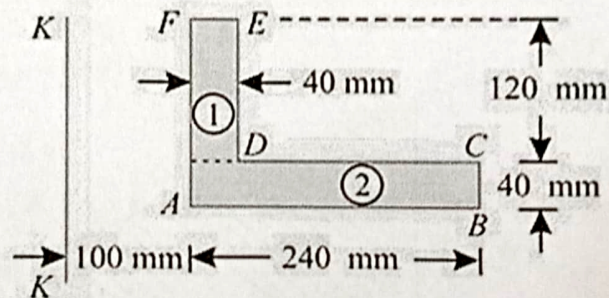


Figure Q. 8 (b)